Factsheet for the finite implication E677 $\models_{\text{fin}}$ E255

May 2026

Abstract

Checked local facts for finite magmas satisfying

$$\text{E677}: \quad x = y \diamond \big(x \diamond ((y \diamond x) \diamond y)\big), \qquad \text{E255}: \quad x = ((x \diamond x) \diamond x) \diamond x.$$

Fix x. Put $a = x \diamond x$, $q = a \diamond x$, and $p = x\backslash x$. The target is

$$\text{E255 at } x \quad \Longleftrightarrow \quad q \diamond x = x.$$

Everything below is proved here from E677 plus finiteness, or marked as external ETP input. No right cancellation, associativity, identity element, or quasigroup structure is used.

1 Setup and finite tools

A magma is a set with one binary operation $\diamond$. For $u \in M$, write

$$L_u(t) = u \diamond t, \qquad R_u(t) = t \diamond u, \qquad S(t) = t \diamond t.$$

If L_u is bijective, $u\backslash v$ denotes the unique w with $u \diamond w = v$. All indices on an L_x-cycle are read modulo its period.

Proposition 1 (Finite left division)**.** *In every finite* E677*-magma, each L_a is bijective and*

$$a\backslash b = b \diamond ((a \diamond b) \diamond a).$$

Also

$$a\backslash(a\backslash b) = (a\backslash b) \diamond (b \diamond a), \qquad b\backslash(a\backslash b) = (a \diamond b) \diamond a.$$

Proof. Proof certificate. E677 gives

$$b = a \diamond \big(b \diamond ((a \diamond b) \diamond a)\big),$$

so L_a is surjective, hence bijective by finiteness. Substitute $a\backslash b$ for b, and use uniqueness of left division. $\square$

Lemma 2 (Two global identities)**.** *For all y, z,*

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y) \tag{T}$$

and

$$(y \diamond (y \diamond z)) \diamond y = z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z)). \tag{K}$$

Proof. Proof certificate. For (T), substitute $y \diamond z$ for the law-variable x in E677, then left-cancel y. For (K), compare (T) with E677 using parameter $y \diamond z$, then left-cancel $y \diamond z$. $\square$

Remark 3 (External ETP guardrails). The unrestricted implication E677 $\models$ E255 is false in the free E677-magma, so any proof must use finiteness. Linear models satisfying E677 satisfy E255; affine-fiber extensions over them are ruled out by ETP structure theory. Finite E677-magmas need not be right-cancellative [3].

2 Local equivalences

Fix a finite E677-magma M and $x \in M$. Put

$$a = x \diamond x, \qquad q = a \diamond x = (x \diamond x) \diamond x, \qquad p = x \backslash x.$$

Proposition 4 (Unique right fixed-point witness)**.** *If $u \diamond x = x$, then $u = q$. Hence*

$$\text{E255 at } x \quad \iff \quad \exists u \; (u \diamond x = x) \quad \iff \quad q \diamond x = x.$$

Proof. Proof certificate. If $u \diamond x = x$, then (T) gives $x = x \diamond (x \diamond u)$. The specialization $(x, y) = (x, x)$ in E677 gives $x = x \diamond (x \diamond q)$. Two left cancellations give $u = q$. $\square$

Proposition 5 (Equivalent local targets)**.** *The following are equivalent:*

$$\text{E255 at } x, \qquad q \diamond x = x, \qquad q \backslash x = x, \qquad ((x \diamond x) \diamond q) \diamond (x \diamond x) = x, \qquad (q \diamond x) \diamond q = p.$$

The fourth identity is the identity sometimes called Lemma L.

Proof. Proof certificate. The first two are Proposition 4. (T), with $(y, z) = (x \diamond x, x)$, gives

$$x = q \diamond (((x \diamond x) \diamond q) \diamond (x \diamond x)).$$

Thus $q \backslash x = ((x \diamond x) \diamond q) \diamond (x \diamond x)$. Finally

$$q \backslash x = x \diamond ((q \diamond x) \diamond q),$$

and left cancellation by x compares this with $p = x \backslash x$. $\square$

Lemma 6 (Fixed maps)**.** *Define*

$$H_x(t) = ((x \diamond t) \diamond x), \qquad F_x(t) = x \diamond (t \diamond x).$$

Then

$$x \backslash t = t \diamond H_x(t), \qquad F_x(L_x(t)) = L_x(H_x(t)).$$

Thus F_x and H_x have the same finite cycle structure. Moreover E255 at x *is equivalent to either map having a fixed point. If fixed points exist, they are unique:*

$$\mathrm{Fix}(F_x) = \{x \backslash q\}, \qquad \mathrm{Fix}(H_x) = \{x \backslash (x \backslash q)\}.$$

Under $q \diamond x \neq x$, both maps are fixed-point-free.

Proof. Proof certificate. E677 with parameter x gives

$$t = x \diamond (t \diamond ((x \diamond t) \diamond x)) = x \diamond (t \diamond H_x(t)),$$

hence $x \backslash t = t \diamond H_x(t)$. Also

$$F_x(L_x(t)) = x \diamond ((x \diamond t) \diamond x) = x \diamond H_x(t) = L_x(H_x(t)),$$

so $F_x = L_x H_x L_x^{-1}$, and F_x, H_x are conjugate.
Suppose $F_x(t) = t$. Comparing

$$x \diamond (t \diamond x) = t = x \diamond (t \diamond H_x(t))$$

and left-cancelling first x and then t, we get $H_x(t) = x$. Thus $(x \diamond t) \diamond x = x$, so Proposition 4 gives $x \diamond t = q$; hence $t = x \backslash q$. Conversely, if $q \diamond x = x$ and $t = x \backslash q$, then $H_x(t) = (x \diamond t) \diamond x = q \diamond x = x$, and the displayed edge formula gives

$$t = x \diamond (t \diamond x) = F_x(t).$$

Thus F_x has a fixed point exactly when E255 at x holds, and then its only fixed point is $x \backslash q$.
The statement for H_x follows by conjugacy: $H_x(s) = s$ iff $F_x(L_x(s)) = L_x(s)$. Hence, when fixed points exist,

$$L_x(s) = x \backslash q, \qquad s = x \backslash (x \backslash q).$$

If $q \diamond x \neq x$, Proposition 4 says that no right fixed point over x exists, so neither F_x nor H_x has a fixed point. □

Lemma 7 (Division return trap)**.** *For all y, z, put*

$$u = y \diamond z, \qquad v = u \diamond y.$$

Then

$$v = z \backslash (y \backslash z), \qquad y \backslash z = z \diamond v. \tag{D}$$

Consequently, if also $z \diamond v = u$, then

$$z = y \diamond u.$$

Proof. Proof certificate. The formula for $y\backslash z$ is just the left-division form of E677:

$$y\backslash z = z \diamond ((y \diamond z) \diamond y) = z \diamond v.$$

Left-dividing this equality by z gives

$$v = z\backslash(y\backslash z).$$

If $z \diamond v = u$, then the displayed identity gives $y\backslash z = u$. Left-multiply by y to obtain $z = y \diamond u$. ☐

3 Orbit facts

Define

$$c_0 = x, \qquad c_{k+1} = x \diamond c_k,$$

and let d be the minimal period of the L_x-orbit of x. Put

$$d_k = c_k \diamond x.$$

Proposition 8 (Basic orbit package)**.** *For all k,*

$$c_{k-1} = c_k \diamond d_{k+1}, \qquad d_1 = c_{d-2},$$

$$x\backslash c_k = c_{k-1}, \qquad c_k\backslash c_{k-1} = d_{k+1}.$$

In particular

$$p = c_{d-1}, \qquad q = x\backslash p = d_1 = c_{d-2} = a \diamond x, \qquad p \diamond a = q, \qquad p\backslash q = a.$$

If $d_j = x$, then $j \equiv d - 2 \pmod{d}$.

Proof. Proof certificate. Applying E677 with parameter x and law-variable c_k gives

$$c_k = x \diamond \big(c_k \diamond ((x \diamond c_k) \diamond x)\big) = x \diamond (c_k \diamond d_{k+1}).$$

Since also $c_k = x \diamond c_{k-1}$, left cancellation by x gives

$$c_{k-1} = c_k \diamond d_{k+1}.$$

The equality $x \diamond q = p$ follows from the specialization

$$x = x \diamond (x \diamond q)$$

of E677 at $(x, y) = (x, x)$, because $p = x\backslash x$. Therefore $q = x\backslash p = c_{d-2}$, while $q = a \diamond x = d_1$ by definition. The division statements follow from $x \diamond c_{k-1} = c_k$ and $c_k \diamond d_{k+1} = c_{k-1}$. In particular $p = c_{d-1}$, $p \diamond a = q$ is the recurrence at $k = d - 1$, and $p\backslash q = a$.

Finally suppose $d_j = x$. By (T) with $(y, z) = (c_j, x)$,

$$x = x \diamond ((c_j \diamond x) \diamond c_j) = x \diamond (x \diamond c_j).$$

Comparing with $x = x \diamond (x \diamond q)$ and left-cancelling x twice gives $c_j = q$. Since $q = c_{d-2}$ on the L_x-cycle, $j \equiv d - 2 \pmod{d}$. ☐

Corollary 9 (Small periods)**.** *If $d = 1$, then* E255 at x *holds. The cases $d = 2$ and $d = 3$ are impossible. Thus a bad point has $d \geq 4$.*

Proof. Proof certificate. $d = 1$ means $x \diamond x = x$, so $q = x$ and $q \diamond x = x$.

If $d = 2$, then $q = c_{d-2} = c_0 = x$, hence $d_1 = q \diamond x = x$, contradicting the forced-index statement in Proposition 8, since $1 \not\equiv 0 \pmod 2$.

Assume $d = 3$. Then

$$c_0 = x, \qquad c_1 = a = q, \qquad c_2 = p,$$

and

$$a \diamond x = a, \qquad x \diamond a = p, \qquad x \diamond p = x, \qquad p \diamond a = a.$$

The orbit recurrence at $k = 1$ gives

$$x = a \diamond (p \diamond x). \tag{D3.1}$$

By (K) with $(y, z) = (x, a)$,

$$a = (x \diamond (x \diamond a)) \diamond x = a \diamond (((x \diamond a) \diamond a) \diamond (x \diamond a)) = a \diamond (a \diamond p).$$

Since also $a \diamond x = a$, left cancellation by a gives

$$a \diamond p = x. \tag{D3.2}$$

Comparing (D3.1) with (D3.2) and left-cancelling a gives

$$p \diamond x = p. \tag{D3.3}$$

Now (K) with $(y, z) = (p, a)$ gives

$$x = (p \diamond (p \diamond a)) \diamond p = a \diamond (((p \diamond a) \diamond a) \diamond (p \diamond a)) = a \diamond ((a \diamond a) \diamond a).$$

Since $a \diamond p = x$, left cancellation by a gives

$$(a \diamond a) \diamond a = p. \tag{D3.4}$$

Finally, (K) with $(y, z) = (a, x)$ gives

$$(a \diamond a) \diamond a = x \diamond (a \diamond a).$$

Using (D3.4) and $x \diamond a = p$, left cancellation by x gives $a \diamond a = a$. Then $(a \diamond a) \diamond a = a$, contradicting (D3.4) and the distinctness of $a = c_1$ and $p = c_2$ on a period-three orbit. $\square$

Lemma 10 (Bad orbit-edge exclusions)**.** *Assume* E255 at x *fails. Then for every k,*

$$d_k = c_k \diamond x \notin \{x, c_{k-1}\},$$

or equivalently

$$c_k \diamond x \neq x, \qquad x \diamond (c_k \diamond x) \neq c_k.$$

Proof. Proof certificate. The first exclusion is Proposition 4: under failure, no element u satisfies $u \diamond x = x$. If $d_k = c_{k-1}$, then

$$H_x(c_{k-1}) = ((x \diamond c_{k-1}) \diamond x) = c_k \diamond x = d_k = c_{k-1},$$

contradicting the fixed-point-free conclusion of Lemma 6. Since $x \diamond c_{k-1} = c_k$, left cancellation by x makes $d_k \neq c_{k-1}$ equivalent to $x \diamond d_k \neq c_k$. □

Lemma 11 (Uniform orbit-edge identities)**.** *For all k,*

$$d_{k+2} = c_k \diamond \big((c_{k+1} \diamond c_k) \diamond c_{k+1}\big), \tag{U1}$$

$$x = d_k \diamond \big((c_k \diamond d_k) \diamond c_k\big), \tag{U2}$$

$$(c_k \diamond d_k) \diamond c_k = x \diamond \big((d_k \diamond x) \diamond d_k\big). \tag{U3}$$

Therefore

$$d_k \backslash x = (c_k \diamond d_k) \diamond c_k = x \diamond \big((d_k \diamond x) \diamond d_k\big).$$

At a bad point,

$$a \backslash x = d_2, \qquad q \backslash x = (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q), \qquad q \backslash a = (p \diamond q) \diamond p.$$

Proof. Proof certificate. (U1) is (K) with $(y, z) = (x, c_k)$. (U2) is (T) with $(y, z) = (c_k, x)$. (U3) is (K) with $(y, z) = (c_k, x)$. The quotient forms read (U2) as left division and use (U3). For the displayed specializations, $a \backslash x = x \diamond ((a \diamond x) \diamond a) = x \diamond (q \diamond a)$, and (U1) at $k = 0$ says $d_2 = x \diamond (q \diamond a)$. Since $d_1 = q$, the general quotient formula at $k = 1$ gives

$$q \backslash x = (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q).$$

Finally $p \diamond a = q$, and (T) with $(y, z) = (p, a)$ gives

$$a = q \diamond ((p \diamond q) \diamond p),$$

so $q \backslash a = (p \diamond q) \diamond p$. □

Proposition 12 (Period-four gate)**.** *Assume $d = 4$. Write*

$$a = c_1, \qquad q = c_2, \qquad p = c_3, \qquad r = q \diamond x, \qquad s = p \diamond x.$$

Then

$$x \diamond a = q, \quad a \diamond x = q, \quad a \diamond r = x, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q,$$

and $r \notin \{x, a, q\}$. Thus $r = p$ or $r \notin \{x, a, q, p\}$. Further,

$$q \diamond s = a, \qquad (p \diamond q) \diamond p = s.$$

If $r = p$, then

$$q \diamond a = q, \qquad s = (q \diamond q) \diamond q = a \diamond (q \diamond q), \qquad s \notin \{x, a, q\},$$

$$H_x(a) = p, \qquad H_x(p) = a.$$

If E255 at x *fails and* $d = 4$*, then either*

$$q \diamond x \notin \{x, a, q, p\},$$

or

$$q \diamond x = p \qquad \textit{and} \qquad p \diamond x \notin \{x, a, q, p\}.$$

In particular, at least one of $q \diamond x$ *and* $p \diamond x$ *lies outside* $\{x, a, q, p\}$*.*

Proof. Proof certificate. The L_x-orbit identities give

$$x \diamond a = q, \qquad x \diamond q = p, \qquad x \diamond p = x.$$

Since $q = d_1 = a \diamond x$, we also have $a \diamond x = q$. The recurrence $c_{k-1} = c_k \diamond d_{k+1}$, for $k = 1, 2, 3$, gives

$$a \diamond r = x, \qquad q \diamond s = a, \qquad p \diamond a = q.$$

We next exclude $r = x, a, q$. If $r = x$, then $a \diamond r = a \diamond x = q$, but $a \diamond r = x$, so $q = x$, impossible on a period-four orbit.

If $r = a$, then (U3) at $k = 0$ gives

$$a = r = x \diamond (q \diamond a).$$

Since $x \diamond x = a$, left cancellation by x gives $q \diamond a = x$. Now (K) with $(y, z) = (q, x)$ gives

$$(q \diamond r) \diamond q = x \diamond ((r \diamond x) \diamond r).$$

With $r = a$, this becomes

$$(q \diamond a) \diamond q = x \diamond ((a \diamond x) \diamond a),$$

hence $p = a$, because the left side is $x \diamond q = p$ and the right side is $x \diamond (q \diamond a) = x \diamond x = a$. This is impossible.

If $r = q$, then $a \diamond q = x$. By (U3) at $k = 1$,

$$(a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q),$$

so $q = x \diamond (q \diamond q)$. Comparing with $q = x \diamond a$ gives $q \diamond q = a$. Applying (K) with $(y, z) = (q, x)$ and using $r = q$ gives

$$(q \diamond q) \diamond q = x \diamond (q \diamond q).$$

The left side is $a \diamond q = x$, while the right side is $x \diamond a = q$, another contradiction. Thus $r \notin \{x, a, q\}$, so either $r = p$ or r is outside the principal orbit $\{x, a, q, p\}$.

The identity $q \diamond s = a$ was obtained above, and

$$(p \diamond q) \diamond p = s$$

is the specialization $q \backslash a = (p \diamond q) \diamond p$ from Lemma 11, because $q \diamond s = a$.

Assume now that $r = p$. From (U3) at $k = 0$,

$$p = r = x \diamond (q \diamond a).$$

Since $x \diamond q = p$, left cancellation by x gives

$$q \diamond a = q. \tag{P4.1}$$

Then (U1) at $k = 1$ gives

$$s = a \diamond ((q \diamond a) \diamond q) = a \diamond (q \diamond q). \tag{P4.2}$$

Also (K) with $(y, z) = (q, a)$, together with (P4.1), gives

$$(q \diamond q) \diamond q = a \diamond ((q \diamond a) \diamond q) = a \diamond (q \diamond q) = s. \tag{P4.3}$$

The exclusions for s are immediate. If $s = x$, then $p \diamond x = x$, so Proposition 4 gives $p = q$. If $s = a$, then

$$a = q \diamond s = q \diamond a = q$$

by (P4.1). If $s = q$, then $p \diamond x = q = p \diamond a$, and left cancellation by p gives $x = a$. Hence $s \notin \{x, a, q\}$.

Finally,

$$H_x(a) = ((x \diamond a) \diamond x) = q \diamond x = r = p, \qquad H_x(p) = ((x \diamond p) \diamond x) = x \diamond x = a.$$

If E255 at x fails, then $r = q \diamond x \neq x$. The preceding exclusions give either $r \notin \{x, a, q, p\}$, in which case we are done, or $r = p$. In the latter case the exclusions above already give $s = p \diamond x \notin \{x, a, q\}$. It remains only to exclude $s = p$. Suppose $s = p$, and put $t = q \diamond q$. Then (P4.2) and (P4.3) give

$$a \diamond t = p, \qquad t \diamond q = p.$$

Also $p \diamond a = q$. Lemma 7, applied with

$$y = a, \qquad z = t, \qquad u = p, \qquad v = q,$$

therefore gives

$$t = a \diamond p = x.$$

But then $p = a \diamond t = a \diamond x = q$, contradicting the distinctness of p and q on the period-four orbit. Hence $s \neq p$, and so $s \notin \{x, a, q, p\}$. □

Proposition 13 (Period-five and period-six first data)**.** *If $d = 5$, write*

$$a = c_1, \quad b = c_2, \quad q = c_3, \quad p = c_4, \quad g = q \diamond x.$$

Then

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q,$$

$$g = a \diamond ((b \diamond a) \diamond b), \quad q \diamond ((a \diamond q) \diamond a) = x, \quad (a \diamond q) \diamond a = x \diamond (g \diamond q),$$

and at a bad point

$$b \diamond x \notin \{x, a\}, \qquad g \notin \{x, b\}, \qquad p \diamond x \notin \{x, q\}.$$

If $d = 6$, write

$$a = c_1, \quad b = c_2, \quad c = c_3, \quad q = c_4, \quad p = c_5, \quad h = c \diamond x, \quad g = q \diamond x.$$

Then

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q,$$
$$h = a \diamond ((b \diamond a) \diamond b), \quad g = b \diamond ((c \diamond b) \diamond c),$$
$$q \diamond ((a \diamond q) \diamond a) = x, \quad (a \diamond q) \diamond a = x \diamond (g \diamond q),$$

and at a bad point

$$b \diamond x \notin \{x, a\}, \qquad h \notin \{x, b\}, \qquad g \notin \{x, c\}, \qquad p \diamond x \notin \{x, q\}.$$

Proof. Proof certificate. Suppose first that $d = 5$. Then $q = c_{d-2} = c_3$ and $p = c_{d-1} = c_4$, so

$$a \diamond x = q, \qquad x \diamond q = p, \qquad x \diamond p = x.$$

The recurrence at $k = 4$ gives $p \diamond a = q$. By (U1) at $k = 1$,

$$g = d_3 = a \diamond ((b \diamond a) \diamond b).$$

By (U2) at $k = 1$,

$$q \diamond ((a \diamond q) \diamond a) = x,$$

and (U3) at $k = 1$ gives

$$(a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q) = x \diamond (g \diamond q).$$

At a bad point, $g = x$ would be $q \diamond x = x$, contrary to the equivalence in Proposition 4; the remaining exclusions are exactly Lemma 10 at $k = 2, 3, 4$.

The $d = 6$ case is the same calculation with $q = c_4$ and $p = c_5$. Thus

$$a \diamond x = q, \qquad x \diamond q = p, \qquad x \diamond p = x, \qquad p \diamond a = q.$$

By (U1) at $k = 1$ and $k = 2$,

$$h = d_3 = a \diamond ((b \diamond a) \diamond b), \qquad g = d_4 = b \diamond ((c \diamond b) \diamond c).$$

By (U2) and (U3) at $k = 1$,

$$q \diamond ((a \diamond q) \diamond a) = x, \qquad (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q) = x \diamond (g \diamond q).$$

At a bad point the displayed exclusions are Lemma 10 at $k = 2, 3, 4, 5$. □

4 Collision and right-tail facts

Proposition 14 (Collision lift)**.** *If $d_i = d_j = e$, then*

$$(c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j.$$

If also $i \not\equiv j \pmod d$, then

$$c_i \diamond e \neq c_j \diamond e.$$

Proof. *Proof certificate.* (T) gives $x = d_k \diamond ((c_k \diamond d_k) \diamond c_k)$. Equal d_i, d_j allow left cancellation by e. If also $c_i \diamond e = c_j \diamond e = s$, then left-cancel s to get $c_i = c_j$. □

Proposition 15 (Right-collision rectangles)**.** *If $y \diamond x = r$, then*

$$(y \diamond r) \diamond y = r \backslash x. \tag{R}$$

Consequently, if $y \diamond x = z \diamond x = r$ and $y \diamond r = z \diamond r$, then $y = z$.

Proof. *Proof certificate.* (T) with $(y, z) = (y, x)$ gives $x = r \diamond ((y \diamond r) \diamond y)$. This is (R). The last claim is left cancellation after applying (R) to y and z. □

Lemma 16 (Right-tail first repetition)**.** *Assume $q \diamond x \neq x$. Define*

$$\rho_0 = x, \qquad \rho_{n+1} = \rho_n \diamond x.$$

Then $\rho_n \neq x$ for $n \geq 1$. Let $1 \leq i < j$ be the first repetition in the tail, with j minimal, and put

$$e = \rho_i = \rho_j, \qquad y = \rho_{i-1}, \qquad z = \rho_{j-1}.$$

Then

$$y \neq z, \qquad y \diamond x = z \diamond x = e,$$

$$(y \diamond e) \diamond y = (z \diamond e) \diamond z = e \backslash x, \qquad y \diamond e \neq z \diamond e.$$

Proof. *Proof certificate.* If some $\rho_n = x$, then $\rho_{n-1} \diamond x = x$, forcing $\rho_{n-1} = q$, contradiction. Finiteness gives a first repetition. Minimality gives $y \neq z$. Proposition 15 gives the rectangle and splitter. □

Proposition 17 (Left-translation labels)**.** *Write $a \diamond b = \sigma(a)(b)$. Then $\sigma(a) \in \mathrm{Sym}(M)$ and the label map $a \mapsto \sigma(a)$ is injective. If $y \diamond x = z \diamond x = r$ with $y \neq z$, then*

$$\sigma(y)(x) = \sigma(z)(x) = r, \qquad \sigma(y)(r) \neq \sigma(z)(r),$$

$$\sigma(\sigma(y)(r))(y) = r \backslash x = \sigma(\sigma(z)(r))(z).$$

Proof. *Proof certificate.* Left translations are permutations by Proposition 1. Suppose $L_y = L_z$. Choose any $x \in M$ and put

$$e = y \diamond x = z \diamond x.$$

The two E677 specializations give

$$x = y \diamond (x \diamond (e \diamond y)), \qquad x = z \diamond (x \diamond (e \diamond z)).$$

Since $L_y = L_z$, the second equality may be written as

$$x = y \diamond (x \diamond (e \diamond z)).$$

Comparing with the first equality and left-cancelling successively by y, then by x, then by e, gives $y = z$. Thus $a \mapsto \sigma(a)$ is injective.

If $y \diamond x = z \diamond x = r$ and $y \neq z$, Proposition 15 gives

$$(y \diamond r) \diamond y = (z \diamond r) \diamond z = r \backslash x.$$

It also says that $y \diamond r = z \diamond r$ would force $y = z$, so $\sigma(y)(r) \neq \sigma(z)(r)$. Rewriting these products in the notation $a \diamond b = \sigma(a)(b)$ gives the displayed formulas. □

5 Bad-point checklist and open gates

Proposition 18 (Bad-point package)**.** *If* E255 at x *fails, then:*

1. *the* L_x*-orbit of* x *has period* $d \geq 4$*;*
2. *no* u *satisfies* $u \diamond x = x$*, so* $x \notin \operatorname{im} R_x$*;*
3. R_x *has a canonical nontrivial collision fiber from Lemma 16;*
4. *on a fiber* $R_x^{-1}(e)$ *as above,* $t \mapsto t \diamond e$ *is injective and obeys (R);*
5. H_x *and* F_x *are fixed-point-free;*
6. *if* $d = 4$*, some one of* $q \diamond x, p \diamond x$ *is outside* $\{x, a, q, p\}$*.*

Proof. Proof certificate. Items: Corollary 9; Proposition 4; Lemma 16; Proposition 15; Lemma 6; Proposition 12. □

Remark 19 (Dependency table for a bad point)*.* The following implications are the current compressed route graph. They are all proved above.

$$\begin{aligned}
\text{E255 at } x \quad &\Longleftrightarrow \quad q \diamond x = x \\
&\Longleftrightarrow \quad \exists u \ (u \diamond x = x) \\
&\Longleftrightarrow \quad q \backslash x = x \\
&\Longleftrightarrow \quad ((x \diamond x) \diamond q) \diamond (x \diamond x) = x \\
&\Longleftrightarrow \quad (q \diamond x) \diamond q = p.
\end{aligned}$$

Under the negation $q \diamond x \neq x$:

$$\begin{aligned}
\text{no } u \diamond x = x \quad &\Longrightarrow \quad x \notin \operatorname{im} R_x \\
&\Longrightarrow \quad R_x \text{ has a nontrivial fiber} \\
&\Longrightarrow \quad y \diamond x = z \diamond x = e, \ y \neq z \\
&\Longrightarrow \quad (y \diamond e) \diamond y = (z \diamond e) \diamond z = e \backslash x, \quad y \diamond e \neq z \diamond e.
\end{aligned}$$

The canonical way to choose the fiber is the first repetition in the right tail $\rho_{n+1} = \rho_n \diamond x$. This removes choice from the splitter route.

For the orbit route:

$$\begin{aligned} q \diamond x \neq x \quad &\Longrightarrow \quad d \geq 4, \\ d = 4 \quad &\Longrightarrow \quad q \diamond x \in \{p\} \cup (M \setminus \{x, a, q, p\}), \\ d = 4,\ q \diamond x = p \quad &\Longrightarrow \quad p \diamond x \notin \{x, a, q, p\}. \end{aligned}$$

So a bad period-four point exits the principal orbit along the right tail by the step $p \diamond x$. No four-element principal-orbit-only bad point survives these facts.

Remark 20 (Finite facts used, nothing else). Only these finite principles appear: finite surjection = bijection; a finite self-map missing a point has a nontrivial fiber; every finite forward orbit repeats; and conjugate finite self-maps have the same cycle structure. Every cancellation in the paper is left cancellation by a proved bijective left translation.

Remark 21 (Do not use).

No right cancellation. No associativity. No identity element.
No universal idempotence. No global "H_x has no cycles".
Do not use $d_k \backslash x = c_{k-2} \diamond c_{k-1}$.
Do not use $p \diamond d_k = d_{k+1}$ unless freshly proved.

Gate 22 (Fixed-point gate)**.** Prove $q \diamond x = x$, equivalently Lemma L or $(q \diamond x) \diamond q = p$, with a visible finite step.

Gate 23 (Splitter gate)**.** Use the canonical right-tail collision fiber, or any collision $d_i = d_j = e$, to turn the injective splitter $t \mapsto t \diamond e$ into either $q \diamond x = x$ or a finite counting contradiction.

Gate 24 (Period-four external gate)**.** At a bad period-four point, feed the forced external element among

$$q \diamond x, \qquad p \diamond x$$

back into the fixed-point gate or the splitter gate.

References

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